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# Structure, energetics and dynamics in crowded amino acid solutions: a molecular dynamics study†

Sibasankar Panigrahy, <sup>‡a</sup> Rahul Sahu, <sup>‡b</sup> Sandeep K. Reddy <sup>b</sup> and  
Divya Nayar <sup>\*a</sup>

A comprehensive understanding of crowding effects on biomolecular processes necessitates investigating the bulk thermodynamic and kinetic properties of the solutions with an accurate molecular representation of the crowded milieu. Recent studies have reparameterized the non-bonded dispersion interaction of solutes to precisely model intermolecular interactions, which would circumvent artificial aggregation as shown by the original force-fields. However, the performance of this reparameterization is yet to be assessed for concentrated crowded solutions in terms of investigating the hydration shell structure, energetics and dynamics. In this study, we perform molecular dynamics simulations of crowded aqueous solutions of five zwitterionic neutral amino acids (Gly, Ala, Thr, Pro, and Ser), mimicking the molecular crowding environment, using a modified AMBER ff99SB-ILDN force-field. We systematically examine and show that the reproducibility of the osmotic coefficients, density, viscosity and self-diffusivity of amino acids improves using the modified force-field in crowded concentrations. The modified force-field also improves the structuring of the solute solvation shells, solute interaction energy and convergence of tails of radial distribution functions, indicating reduction in the artificial aggregation. Our results also indicate that the hydrogen bonding network of water weakens and water molecules anomalously diffuse at small time scales in the crowded solutions. These results underscore the significance of examining the solution properties and anomalous hydration behaviour of water in crowded solutions, which have implications in shaping the structure and dynamics of biomolecules. The findings also illustrate the improvement in predicting bulk solution properties using the modified force-field, thereby providing an approach towards accurate modeling of crowded molecular solutions.

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## Introduction

Understanding the consequences of a realistic intracellular environment on biomolecular structure, dynamics and function still remains a challenge *in vitro* as well as *in silico*.<sup>1–3</sup> The interior of a living cell is packed with high concentrations (typically 200 to 400 g L<sup>−1</sup>) of metabolites and macromolecules such as proteins, and lipids. Such a tightly packed milieu exerts volume exclusion effects<sup>4</sup> and imparts non-ideal solution behaviour leading to enhanced thermodynamic activity<sup>5</sup> and reduced dynamics<sup>6–8</sup> of biomolecules. A number of experimental studies have attempted to examine crowding effects,

but have been restricted with the challenge of achieving sufficient resolution in highly concentrated solutions.<sup>9–11</sup> With the recent advances in computational resources, there is a considerable interest in using computer simulations to examine biomolecular processes in a crowded environment. One of the major challenges in molecular simulations, however, is achieving accurate description of the crowded environment at the molecular level.

Several simulation studies have primarily adopted simplified models to represent the crowded environment and the solvent,<sup>1,2,12,13</sup> or have adopted multi-scale approaches such as those that combine a coarse-grained description of proteins with a lattice-based description of solvent-mediated hydrodynamic effects,<sup>14–17</sup> either due to high computational costs associated with simulating atomistic models or due to the lack of availability of high resolution atomistic models that would accurately mimic the intermolecular interactions.<sup>1</sup> Moreover, such a simplified description of crowders is based on the assumption that crowding effects are exerted only due to the

<sup>a</sup> Department of Materials Science and Engineering, Indian Institute of Technology Delhi, New Delhi 110016, India. E-mail: divyanayar@mse.iitd.ac.in

<sup>b</sup> Center for Computational and Data Sciences, Indian Institute of Technology Kharagpur, West Bengal 721302, India

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\* These authors contributed equally to this work.

volume exclusion (steric) effects of the large-sized crowders with little or no role of the soft crowder intermolecular interactions.<sup>4,18</sup> Recent studies, however, have pointed out that the soft attractive intermolecular interactions in a crowded environment play a crucial role in regulating biomolecular processes.<sup>19–22</sup> Therefore, it is critical to have an accurate atomistic representation of the crowded environment accounting for the chemical nature (intermolecular interactions) of the crowders, capable of mimicking the thermodynamic and kinetic features of a real crowded environment. This necessitates accurate modeling of intermolecular interactions such as of inter-chain interactions between macromolecules in such complex systems, which remains a challenge to be addressed in atomistic simulations.

Interestingly, longer time-scale molecular dynamics simulations using all-atom biomolecular force-fields such as CHARMM<sup>23</sup> and AMBER<sup>24,25</sup> have highlighted an overestimation of the attractive solute–solute interactions resulting in an artificial aggregation in the systems of small peptides,<sup>26</sup> self-interacting dipeptide systems,<sup>27–29</sup> denatured proteins and assemblies.<sup>30–32</sup> Such artificial aggregation is expected to be enhanced in crowded packing fractions of up to 20–30% where multi-component interactions tend to be significant. The improper balance of solute–solute and solute–solvent interactions has also been observed while examining protein stability in aqueous solutions of competing osmolytes, such as of betaine–water interactions in aqueous betaine–TMAO solutions<sup>33</sup> or TMAO–water interactions in aqueous urea–TMAO solutions.<sup>34</sup> To correct for the overly attractive solute–solute interactions, the literature suggests two plausible routes. However, there is still a lack of agreement on the best route to be adopted. One possibility is that the solute–water interactions can be improved with the available water models.<sup>30,35</sup> It has been pointed out that the current rigid-body water models, such as the TIPnP family, underestimate the solute–water interactions and even predict overly compact unfolded or disordered ensembles for proteins.<sup>32,36,37</sup> In this direction, new water models have been developed such as TIP4P-D with improved water dispersion interactions<sup>31</sup> and an OPC model with optimized distribution of point charges.<sup>38</sup> The improved parameterization, however, affects and poses the challenge of producing stable folded ensembles of proteins.<sup>39</sup> An alternative second possible route would be to improve the solute–solute non-bonded interactions as was suggested by Yoo *et al.*<sup>40</sup> They have proposed incrementing the size parameter of mixed Lennard-Jones interaction between the charged amine and the carboxyl groups of amino acid termini to reduce the artificial aggregation of proteins. They showed that this modification does not affect the solvation free energies or the bonded interactions. A few prior studies have assessed the performance of this reparameterization on different force-fields to estimate osmotic coefficients of zwitterionic amino acid solutions at low concentrations.<sup>41,42</sup> However, the applicability of this reparameterization strategy is yet to be examined for computing the bulk thermodynamic and kinetic properties of high concentrations (crowded packing fractions) of solutions where solute–solute attractions are expected to be dominant.

To model proteins as macromolecular crowders accurately, it is important to first examine the crowded environment of amino acids that are building blocks of proteins, essentially representing molecular crowders. In this work, we apply the reparameterization strategy of Yoo *et al.* for computing the bulk properties of crowded amino acid solutions using the modified force-field model of AMBER ff99SB-ILDN,<sup>25</sup> herein denoted as “NBFIX” reparameterization. We systematically compute properties such as the osmotic coefficients, viscosity and self-diffusivity coefficient of five neutral amino acid (glycine or Gly, alanine or Ala, serine or Ser, threonine or Thr, and proline or Pro) aqueous solutions at crowded concentrations using molecular dynamics simulations and compare the results with experimental values. We also quantitatively characterize the structure and energetics of solvation along with the solvation behaviour of water in these crowded solutions. The results demonstrate that the modified force-field performs well in terms of improving the agreement between experimental and simulated osmotic coefficients in crowded concentrations. The reparameterization strategy also improves the concentration-dependence of dynamic properties such as viscosity and self-diffusivity coefficients of solutes. The results also show that the reparameterization improves the structure of the solvation shells and the associated energetics of the amino acids, indicating reduction in the overly attractive hydrophobic or charged solute–solute interactions. We also examine the tetrahedral order and diffusion of water in these solutions. Both the order and diffusion of water decrease in crowded solutions, with anomalous diffusion of water observed in the small time scale regime.

## Computational details

### System setup

Molecular dynamics simulations were performed using GROMACS package version 2019.3.<sup>43</sup> Based on the availability of the experimental data, aqueous solutions of five zwitterionic neutral amino acids have been examined *i.e.* glycine (Gly), non-polar residues alanine (Ala) and proline (Pro), and polar residues serine (Ser) and threonine (Thr) at 2 M, 2.5 M, and 3 M corresponding to high packing fractions (see Table 1). AMBER ff99SB-ILDN<sup>25,44</sup> along with the TIP3P water model<sup>45</sup> was used to model all the amino acid solutions. GROMACS uses the simplest form of Lennard-Jones (LJ) potential function for AMBER force fields given by,

$$V_{\text{LJ}}(r_{ij}) = 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \quad (1)$$

where  $r_{ij}$  is the separation distance for two like or unlike atoms,  $\epsilon_{ij}$  is the depth of the attractive well, and  $\sigma_{ij}$  is the collision diameter. AMBER uses the Lorentz–Berthelot (LB) mixing rule to describe LJ interaction between two unlike atoms. According to the LB rule,  $\epsilon_{ij}$  and  $\sigma_{ij}$  parameters for two unlike atoms  $i$  and  $j$  are defined by

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \quad (2)$$

**Table 1** Details on the number of amino acid molecules ( $N_{\text{solute}}$ ), the number of water molecules ( $N_{\text{water}}$ ), the corresponding packing fractions of the solutes and the apparent molar volume ( $V_m$ )<sup>48</sup> of aqueous solutions are reported. GROMACS utility *freevolume*<sup>49</sup> was used to calculate the molecular packing fraction

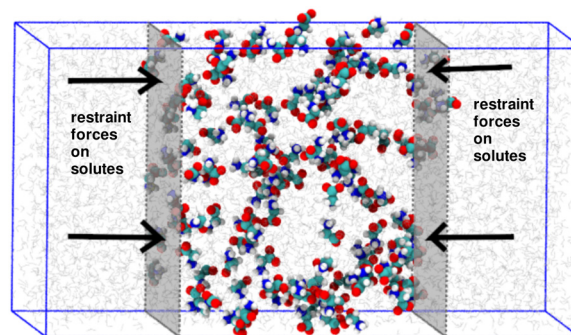
| System  | Concentration (M) | $N_{\text{solute}}$ | $N_{\text{water}}$ | Solute packing fraction | $V_m$ (cm <sup>3</sup> mol <sup>-1</sup> ) |
|---------|-------------------|---------------------|--------------------|-------------------------|--------------------------------------------|
| Gly     | 2                 | 260                 | 5945               | 0.104                   | 41.98                                      |
|         | 2.5               | 325                 | 5701               | 0.129                   | 41.04                                      |
|         | 3                 | 390                 | 5441               | 0.159                   | 40.17                                      |
| Ala     | 2                 | 260                 | 5668               | 0.132                   | 61.64                                      |
|         | 2.5               | 325                 | 5330               | 0.164                   | 60.53                                      |
|         | 3                 | 390                 | 5036               | 0.198                   | 59.82                                      |
| Ser     | 2                 | 260                 | 5585               | 0.141                   | 60.43                                      |
|         | 2.5               | 325                 | 5221               | 0.179                   | 59.54                                      |
|         | 3                 | 390                 | 4924               | 0.215                   | 58.92                                      |
| 3*Pro   | 2                 | 260                 | 5413               | 0.165                   | 78.91                                      |
|         | 2.5               | 325                 | 4999               | 0.210                   | 77.64                                      |
|         | 3                 | 390                 | 4609               | 0.250                   | 76.70                                      |
| Thr     | 2                 | 260                 | 5393               | 0.167                   | 75.71                                      |
|         | 2.5               | 325                 | 5064               | 0.210                   | 75.06                                      |
|         | 3                 | 390                 | 4667               | 0.253                   | 74.26                                      |
| Gly-Gly | 1                 | 130                 | 6142               | 0.096                   | 97.49                                      |
|         | 1.26              | 163                 | 5926               | 0.124                   | 97.08                                      |
|         | 1.49              | 195                 | 5695               | 0.148                   | 95.75                                      |

$$\sigma_{ij} = (\sigma_i + \sigma_j)/2.$$

Yoo *et al.* proposed to modify the mixed size ( $\sigma_{ij}$ ) parameter of the interactions between the amine nitrogen and carboxylate oxygen (N:O) of Gly, Ala, Ser, Thr, and Pro by increasing the  $\sigma_{ij}$  value by  $\Delta\sigma = 0.08$  Å.<sup>40</sup> Therefore the new atom pair size can be defined as  $\sigma_{ij}^{\text{new}} = \sigma_{ij} + \Delta\sigma$ . Additionally, the same size increment parameter ( $\Delta\sigma$ ) was also used for the mixed intermolecular interactions between the hydroxyl oxygen (OH) and amino nitrogen (OH:N), carboxylate oxygen (OH:O), and itself (OH:OH) in the Ser and Thr systems to correct the respective simulated osmotic coefficients.<sup>41</sup> The values of  $\sigma_{ij}^{\text{new}}$  used are summarized in Table S1 (ESI<sup>†</sup>). Such a unified correction was calibrated and proposed at only 2 M of the amino acid solutions. To assess how the modification would perform at higher concentrations, we have examined the effect of a range of  $\Delta\sigma$  parameters with values of 0.06, 0.07, 0.08, 0.09, and 0.10 Å and conducted osmotic pressure simulations of all the amino acids corresponding to each of the modified parameterizations.

### Osmotic pressure simulations

Roux's group<sup>46</sup> used MD simulations to calculate the osmotic coefficient ( $\phi$ ) of the electrolyte solutions (NaCl, KCl) and reproduced the experimental osmotic pressure by modifying the Lennard-Jones (LJ) size parameter ( $\sigma$ ) of the mixed ion-ion interactions. The method involved measuring the pressure exerted by "virtual" walls on the ions and the resultant computed osmotic coefficient was in excellent agreement with the experimental value. The procedure was later followed by other groups for computing the osmotic coefficients.<sup>47</sup> Initially, the required number of amino acids (Table 1) were placed randomly (using *insert-molecules* implemented by GROMACS) in a cubic box of side length 48 Å. The simulation box was then



**Fig. 1** A simulation setup for osmotic pressure and coefficient computation. The central region of the box consists of the amino acid molecules and water (grey). The grey colored walls represent the virtual walls. The color scheme for molecules is carbon (cyan), nitrogen (blue), hydrogen (white), and oxygen (red).

elongated by 24 Å on either side of the z-axis and solvated with water, giving it final box dimensions of  $48 \times 48 \times 96$  Å. The *gmx insert-molecule* tool allows the solute molecules to remain restricted within the volume of  $48 \times 48 \times 48$  Å. This results in three compartments where the aqueous solution of the amino acid compartment lies between the two pure water compartments (Fig. 1), separated by two *xy*-aligned virtual semipermeable membranes. The boundaries (*xy*-planes) of the aqueous amino acid solution chamber act as virtual walls. To maintain the virtual walls, an external flat-bottomed restraint potential (FBP) was applied using a force constant of  $1000 \text{ kJ mol}^{-1} \text{ nm}^{-2}$  along the *z*-direction to the solute heavy atoms so that only water molecules can pass through the virtual walls, freely without any bias, situated at  $z = z_0 \pm 24$  Å, where  $z_0$  is the center of the *z*-axis of the cuboid box, *i.e.*, an energy penalty was imposed on the motion of the solute molecules if they cross the virtual walls.

The energy of the system was minimized using the steepest descent algorithm. Following this, each system was gradually heated to 298 K over a 350 ps time period in an *NVT* ensemble with a velocity rescale thermostat. Prior to the 100 ns production run, the systems were equilibrated in an *NPT* ensemble for a 1 ns time scale using the Nosé-Hoover thermostat and Parrinello-Rahman barostat, while the temperature and pressure were maintained at 298 K and 1 bar, respectively. The simulation box was coupled using a semi-isotropic pressure coupling so that changes in the simulation box size in the *z*-axis could occur independently of changes in the *x*- and *y*-axes. A leap-frog algorithm was used to integrate the equation of motion with a time step of 2.5 fs. The cut-off for computing the van der Waals interactions was used as 10 Å. For electrostatic energy computation, the particle mesh Ewald algorithm<sup>50</sup> was used with the real-space cut-off as 10 Å and Fourier grid spacing of 1.8 Å. We modified the GROMACS source code to store the forces exerted by the FBP at every time step. The force was averaged over every 20 ns block over a 100 ns trajectory. Osmotic pressure ( $\pi$ ) was then calculated by dividing the average force by the surface area of the "virtual" semipermeable membrane and the osmotic coefficient ( $\phi$ ) was evaluated

according to the equation:

$$\varphi = \frac{\pi \cdot V_W}{R \cdot T \cdot m \cdot \nu \cdot M_W} \quad (3)$$

where  $V_W$  is the partial molal volume of water (taken to be  $0.018 \text{ L mol}^{-1}$ ),  $R$  is the gas constant,  $T$  is the temperature,  $m$  is the molality,  $M_W$  is the molecular weight of water ( $0.018013 \text{ kg mol}^{-1}$ ), and  $\nu$  is the vant Hoff factor (1 for non-electrolytes). The experimental data for the osmotic coefficients have been taken from references as summarized in Table S2 (ESI<sup>†</sup>) and have been extrapolated, wherever necessary, to obtain high concentration data.<sup>47</sup>

### Bulk simulations

For examining the effects of force field modifications on the thermodynamic and kinetic properties of the peptide systems, we carried out MD simulations of the above-mentioned five amino acids at different concentrations using modified force field parameters. Initially, for each system, a simulation box of size  $60 \times 60 \times 60 \text{ \AA}^3$  was packed with amino acids and solvated with water. The energy of each system was minimized prior to the incremental heating to reach the temperature 298 K in the *NVT* ensemble using a velocity rescale thermostat. Then the systems were equilibrated for 1 ns followed by a production run for 100 ns in the *NPT* ensemble. During equilibration and the production run, the temperature and pressure were maintained at 298 K using a Nosé–Hoover thermostat and at 1 bar using a Parrinello–Rahman barostat, respectively. Other simulation parameters were the same as in the case of osmotic pressure simulations. System trajectories were stored every 1 ps for analysis.

### Computation of observables

We calculated various system properties, such as density, viscosity, diffusion coefficients of solute molecules, radial distribution functions, solute interaction energy and tetrahedral order of water. We used GROMACS utility “gmX energy” to compute the bulk densities of the peptide solutions from the 100 ns production run trajectories. The error estimate is based on block averages over five blocks or splits of the trajectory. Computed densities were plotted against the molal concentration of the solute and fitted to the quadratic equation,  $\rho(c) = \rho_0 + \rho_{\text{incr}}c + \beta c^2$  where  $\rho(c)$  is the solute density at the molal concentration  $c$ ,  $\rho_0$  is the density of the system in the absence of the solute, and  $\rho_{\text{incr}}$  and  $\beta$  are the fitted parameters. The linear coefficient  $\rho_{\text{incr}}$  is commonly defined as the density increment.<sup>51</sup> The  $\rho_{\text{incr}}$  was scaled by a scaling factor defined as the ratio of the density of pure water and of the TIP3P water model,  $\rho_{\text{expt}}^{\text{water}}/\rho_{\text{sim}}$ , to account for the errors of the water model.<sup>51</sup>

The shear viscosity of the solute molecules in the solution was computed by splitting the trajectory into 10 splits of 10 ns each. Ten separate simulations were performed each for 1 ns starting after each 10 ns block of the original 100 ns trajectory, *i.e.* at 10, 20, 30, 40 ns, *etc.* The velocities were frequently stored every 10 fs in these simulations. The transverse current

autocorrelation functions were computed using the “gmX tcdf” utility. The averaged  $k$ -dependent viscosities were then fitted as a function of  $k$  using the equation  $f = y_0(1 - ak^2)$ , where  $y_0$  is the shear viscosity and  $a$  is the regression coefficient. To estimate the associated error, the  $k$ -vectors obtained from each autocorrelation functions were fitted to the aforementioned equation to evaluate the shear viscosity of each block of trajectory and the standard deviation was then computed. The computed shear viscosities were plotted as a function of molar concentration of solute and fitted to the equation  $\eta(c) = \eta_0 + \eta_{\text{incr}}c + \beta c^2$ , where  $\eta(c)$  is the solute viscosity at molar concentration  $c$ ,  $\eta_0$  is the viscosity in the absence of solute (pure water), and  $\eta_{\text{incr}}$  and  $\beta$  are the regression coefficients. The linear coefficient  $\eta_{\text{incr}}$  is known as the viscosity increment. The  $\eta_{\text{incr}}$  was scaled by a scaling factor defined by the ratio of viscosity of pure water and of TIP3P water,  $\eta_{\text{expt}}^{\text{water}}/\eta_{\text{sim}}$ , to account for the errors of the water model.<sup>51</sup> The shear viscosity of water has been shown to exhibit no system-size dependence.<sup>52</sup> Therefore, no finite-size correction was applied while computing the shear viscosities.

The self-diffusivity coefficients ( $D$ ) were calculated using mean-squared displacement (msd) of the center of mass of amino acid residues or center of mass of water molecules,

$$6Dt = \text{msd}(t) = \left\langle |\mathbf{r}_{\text{com}}(t' + t) - \mathbf{r}_{\text{com}}(t')|^2 \right\rangle_{t'} \quad (4)$$

where  $\mathbf{r}_{\text{com}}$  symbolizes the position of the center of mass of amino acid residues or water molecules. The finite-size effects on the diffusion coefficient in the periodic cubic box were corrected using the equation:

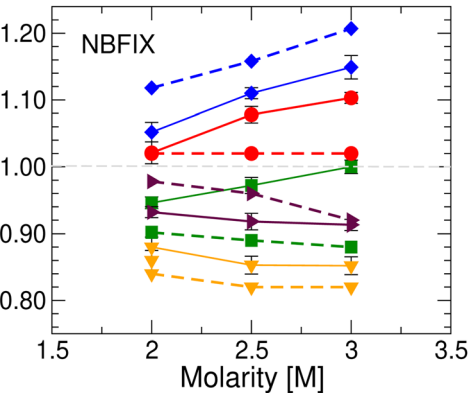
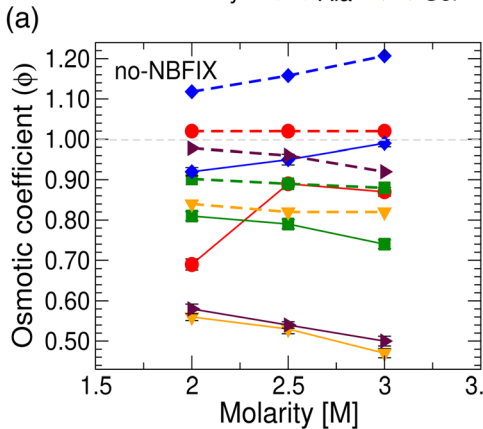
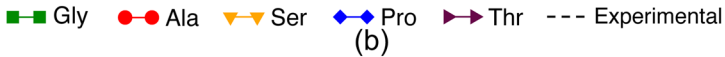
$$D = D_{\text{PBC}} + \frac{k_B T \xi}{6\pi\eta L} \quad (5)$$

Here,  $D_{\text{PBC}}$  is the diffusion coefficient obtained from the simulation and the correction factor  $\frac{k_B T \xi}{6\pi\eta L}$  was added to it, where  $k_B$  is the Boltzmann constant,  $T$  is the temperature,  $L$  is the box length,  $\eta$  is the shear viscosity of pure water, and the correction factor  $\xi = 2.83729$ .<sup>52</sup>

The computed  $D$  values were plotted as a function of molar concentration of solute and fitted to the equation  $D(c) = D_0 + D_{\text{decr}}c$ , where  $D(c)$  is the diffusivity coefficient at the molar concentration  $c$ ,  $D_0$  is the diffusion coefficient in the absence of solute, and  $D_{\text{decr}}$  is the linear coefficient known as the diffusivity coefficient decrement.

Radial distribution functions (rdfs) for the various types of intermolecular interactions involved with amino nitrogen and carboxylate oxygen (N–O), amino nitrogen and oxygen of water (N–OW), and carboxylate oxygen and hydrogen of water (O–HW) were calculated from the last 20 ns trajectories of the production run.

The energy of interactions were computed between the pair of functional groups such as of amine and carboxylate groups ( $\text{NH}_3^+ - \text{COO}^-$ ) of zwitterions, of amine group and water ( $\text{NH}_3^+ - \text{water}$ ), and of carboxylate group and water ( $\text{COO}^- - \text{water}$ ). In order to understand the effect of reparametrization on the local solvation shell energetics, the solute–solute and solute–water



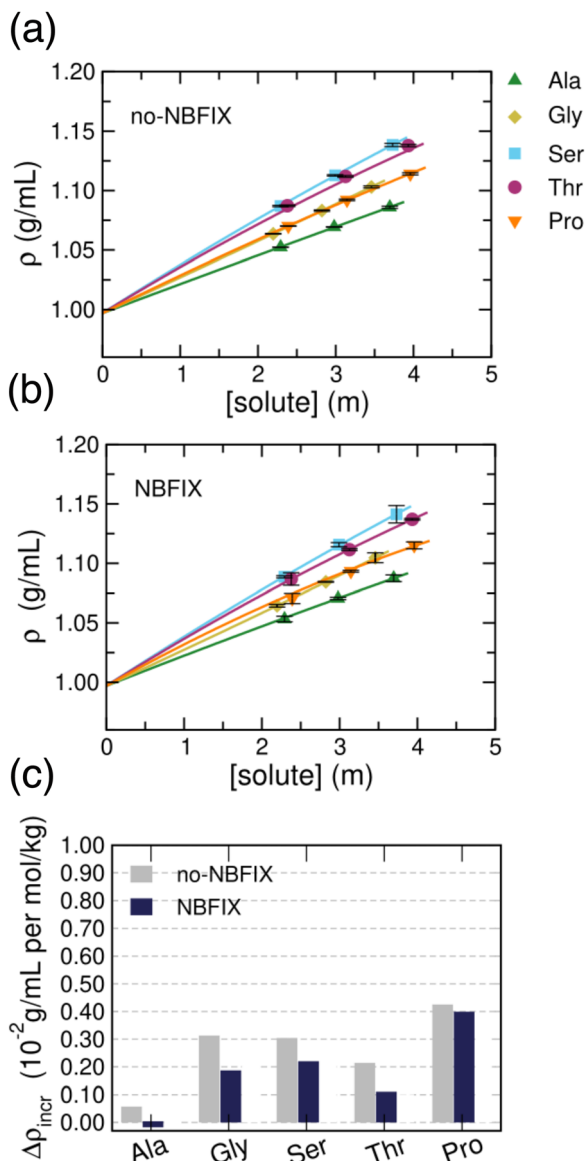


Fig. 3 Density of aqueous amino acid solutions as a function of various amino acid concentrations when (a) original (no-NBFI) and (b) NBFI parameterization are used. Part (c) shows the deviation of simulated scaled density increment ( $\Delta\rho_{\text{incr}}$ ) values from experimental values (see Table S2, ESI†) defined by  $\rho_{\text{incr}}(\text{expt}) - \rho_{\text{incr}}(\text{simulated})$  for different amino acid aqueous solutions.

except for Ala, in all the cases when original or NBFI parameterization is applied. Fig. 3(c) shows that on applying NBFI parameterization, the deviation from experimental values ( $\Delta\rho_{\text{incr}}$ ) decreases for all the amino acids. In particular for Ala, the difference almost vanishes. However, it can be seen that the absolute difference in ( $\Delta\rho_{\text{incr}}$ ) between no-NBFI and NBFI parameterization is not significantly high (around 6–10%) for all the amino acids. In order to compare between the two kinds of parameterization, we computed the mean absolute deviation (MAD) values using eqn (S1), (ESI†). The values obtained are 0.0036 and 0.0019 g mL<sup>-1</sup> for no-NBFI and NBFI parameterization, respectively, which are not significantly large.

Therefore, the original force-field performs well in reproducing the experimental densities even at high, crowded concentrations and NBFI modification may not be necessary to use for reproducing experimental densities of these solutions. It is to be noted, however, that the simulated apparent molar volume ( $V_{\text{m}}$ ) values of the solutions (see Table 1) are slightly underestimated when compared with the experimental values. For instance, on comparison with the experimental  $V_{\text{m}}$  values for Gly,<sup>55</sup> Ser, and Thr<sup>56</sup> solutions, the differences of  $\approx 3$ , 7 and 9 cm<sup>3</sup> mol<sup>-1</sup> are observed, respectively.

### Viscosity

We next examine the performance of NBFI parameterization for the viscosity of the aqueous solutions at the three concentrations.

Fig. 4(a) and (b) show that viscosity increases with the concentration of amino acids when the original and NBFI parameterization are applied. The viscosity of amino acids decreases in the order Thr, Pro, Ser, Ala, and Gly at all concentrations, irrespective of whether the original or reparameterized force-field is used. High viscosity of hydrophilic residues (Ser, Thr) is expected since the OH side-chain groups can induce additional attractive solute–solute interactions leading to high viscosity. The high viscosity of Pro may be attributed to the sterically hindered ring structure of Pro, which reduces the mobility of solutes and increases the viscosity. The small-sized hydrophobic residues (Ala, Gly) show low viscosity values as compared to the others. Fig. 4(c) shows the deviation of the simulated scaled viscosity increments from the experimental values ( $\Delta\eta_{\text{incr}}$ ) for different amino acid solutions when original and NBFI parameterization are used. It can be seen that the simulated viscosity increment is lower than that in experiments for Ala, Pro and Thr when either original or NBFI parameterization is used (see Fig. S4(b), ESI†). The NBFI parameterization performs better than the original parameterization in reducing the  $\Delta\eta_{\text{incr}}$  for Gly, Ala, Pro, and Ser. Only for Thr, the original force-field parameters predict the scaled viscosity increments closer to the experiment (Fig. S4(b), ESI†). On computing the overall mean absolute deviation of  $\Delta\eta_{\text{incr}}$ , we obtain the values as 0.059 and 0.021 mPa s M<sup>-1</sup> for no-NBFI and NBFI parameterization, respectively. It can be seen that the NBFI reparameterization shows the least overall deviation in  $\Delta\eta_{\text{incr}}$ . Therefore, it can be concluded that the NBFI parameterization improves the agreement between the experimental and simulated concentration-dependence of viscosity.

### Diffusivity

The next dynamic property that we compute is the self-diffusivity coefficient ( $D$ ) of the amino acids in their aqueous solutions at the three concentrations. Fig. 5(a) and (b) show the variation in the simulated diffusivity coefficient ( $D$ ) with amino acid concentration using the original and NBFI parameterization, respectively. As expected, the  $D$  decreases for all the amino acids as their concentration increases. The order in which the amino acids show diffusivity values is Gly > Ala > Ser > Pro > Thr. This is the opposite trend to that observed for viscosity

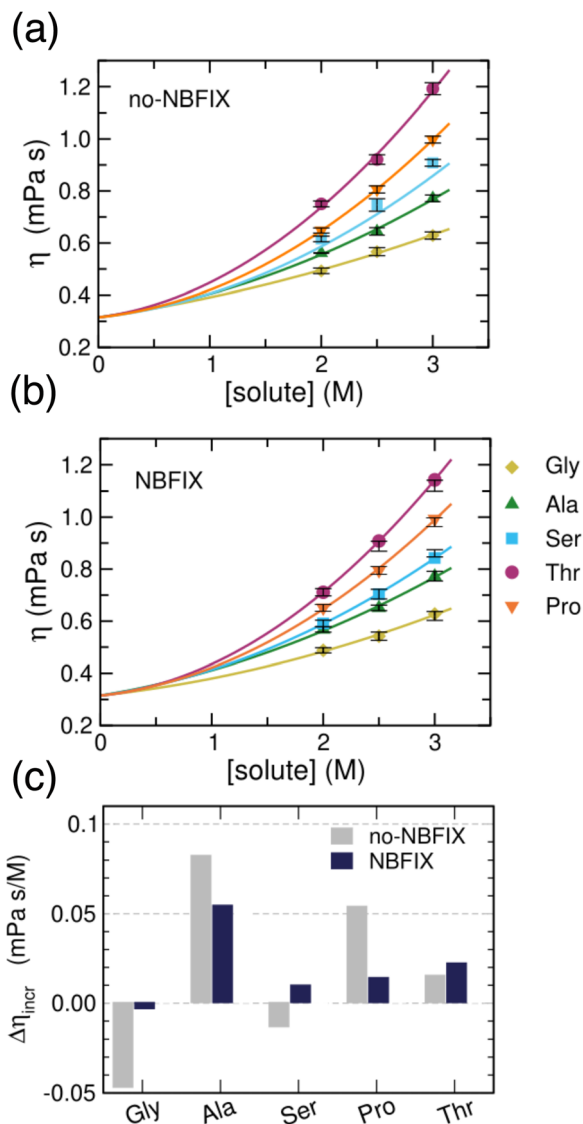


Fig. 4 Viscosity of aqueous amino acid solutions as a function of various amino acid concentrations when (a) original (no-NBFX) and (b) NBFX parameterization are used. Part (c) shows the deviation of the simulated scaled viscosity increments ( $\Delta\eta_{\text{incr}}$ ) from the experimental values (see Table S2, ESI†) defined by  $\eta_{\text{incr}}(\text{expt}) - \eta_{\text{incr}}(\text{simulated})$  for different amino acid aqueous solutions.

(Fig. 4) where Thr shows the highest and Gly shows the least viscosity of the solution.

Fig. 5(c) shows the deviation of the simulated diffusivity decrement from experimental values ( $\Delta D_{\text{decr}}$ ). It can be seen that the diffusivity decrement is overestimated using the original force-field for all the amino acids except Ser when compared with experimental values (see Fig. S4(c), ESI†). The deviation from experimental values is large even when the NBFX parameters are used as indicated by the large  $\Delta D_{\text{decr}}$ . However, the mean absolute deviation (MAD) values are 2.01 and 1.88 ( $\times 10^{-6} \text{ cm}^2 \text{ s}^{-1} \text{ M}^{-1}$ ) for the no-NBFX and NBFX parameterizations, respectively. Therefore, in comparison to no-NBFX, the performance of NBFX is slightly better in reproducing the concentration-dependence of the diffusivity

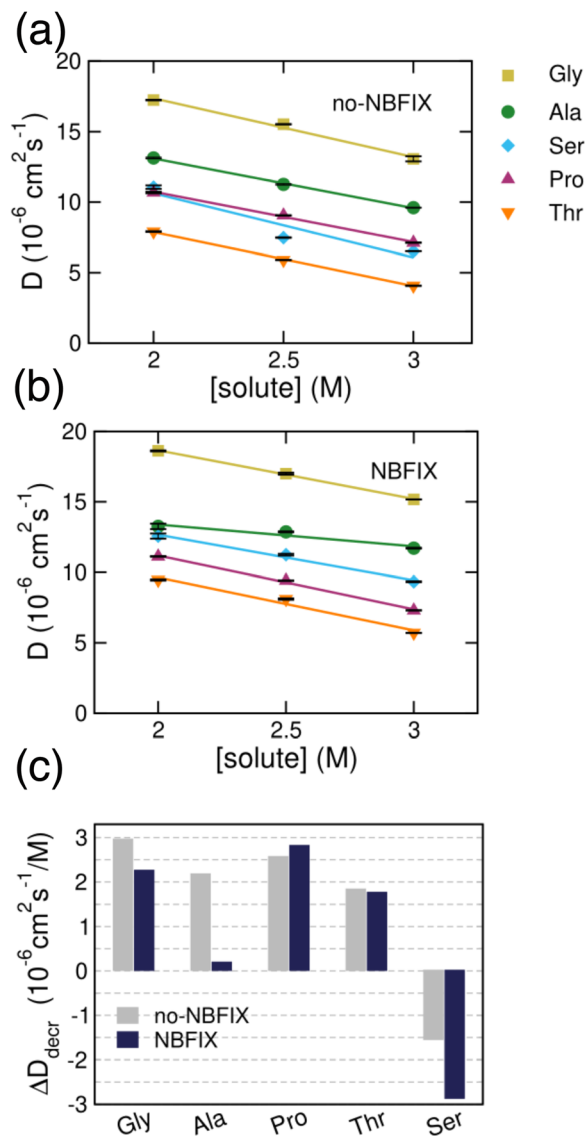


Fig. 5 Self-diffusivity coefficients of amino acids as a function of varying amino acid concentration when (a) original (no-NBFX) and (b) NBFX parameterization are used. Part (c) shows the deviation of absolute decrement in diffusion coefficient ( $\Delta D_{\text{decr}}$ ) from experimental values (see Table S2, ESI†) defined by  $D_{\text{decr}}(\text{expt}) - D_{\text{decr}}(\text{simulated})$  for different amino acid aqueous solutions.

coefficient. Therefore, based on the results obtained for osmotic coefficient, density, diffusivity and viscosity, we can conclude that the NBFX parameterization is successful in reducing the overly attractive solute-solute interactions and therefore performs well in either reproducing the experimental values or the experimental solute concentration-dependence of these properties. We now assess the effect of the NBFX reparameterization for examining the structuring, interaction energy and dynamics of water in the solute solvation shells.

#### Hydration structure and dynamics of aqueous crowded solutions

Fig. 6 shows the radial distribution functions (rdfs) for the termini-termini ( $g(r_{\text{N-O}})$ ) and termini-water ( $g(r_{\text{N-OW}})$ )

interactions of Ser (and other amino acids in Fig. S5, S7 and S8, ESI†) at three concentrations. The termini–termini interactions have quite high probability when the original force-field parameters are used, implying the tendency of solutes to cluster at such high concentrations. On applying the NBFIX parameterization, the termini–termini interactions decrease as indicated by the decrease in the first peak height in the  $g(r_{\text{N-O}})$  at all three concentrations of all the amino acid solutions. Fig. 6 indicates that the hydrophilic residues, Ser and Thr (Fig. S5, ESI†), show the largest first peak height in  $g(r_{\text{N-O}})$  when the original force-field is used. This may be due to the OH groups present in the side-chain that bring the solutes closer due to favourable side-chain interactions. The termini–water interactions, on the other hand, become more favourable on applying NBFIX as indicated by the increase in the first peak height in the  $g(r_{\text{N-OW}})$  or  $g(r_{\text{O-HW}})$  (also see Fig. S7 and S8, ESI†). This indicates that to compensate for the decrease in the solute–solute interactions, there is an increase in the solute–water interactions, which is reflected in the increasing first peak height. Therefore, this highlights that the amino acid–amino acid association decreases and there is increased fraction of “free” amino acid molecules that are fully hydrated. Moreover, the tails of the rdfs converge to 1 when the NBFIX reparameterization is applied as compared to when the original force-field is used, indicating reduction in the aggregate formation, as also quantified by distributions of clusters<sup>57–60</sup> as a function of cluster sizes in Fig. S6 (ESI†). Such an improvement in the convergence of the tails is more predominant in the hydrophilic amino acid (Ser and Thr) solutions (see Fig. S5, S7 and S8, ESI†).

Therefore, for all the amino acid solutions, there is a strong indication of reduction in the over-attractive solute–solute interactions, which causes reduction in the finite-size effects, which are often artifacts of aqueous solutions of such solutes at high concentrations. The improvement in rdf tail convergence, especially for the hydrophilic amino acids, would help in accurate prediction of other thermodynamic properties such as activity and activity coefficients that can be computed using Kirkwood–Buff integrals.<sup>61</sup>

Fig. 7 shows the energy of intermolecular interactions between the  $\text{NH}_3^+ - \text{COO}^-$  (first column),  $\text{NH}_3^+ - \text{water}$  (second column) and  $\text{COO}^- - \text{water}$  (third column) for Gly, Ser and Pro solutions as a function of concentration. It can be seen that the termini–termini interaction is highly favourable (more negative) when the original force-field is used. On applying the NBFIX parameterization the solute–solute energy becomes less favourable (less negative). Consequently, the solute termini ( $\text{NH}_3^+$  or  $\text{COO}^-$ )–water interaction energy is seen to become favourable (more negative) when NBFIX is applied as compared to that obtained using the original force-field. Similar trends are observed for Ala and Thr energy of interactions as shown in Fig. S9 (ESI†).

Gly and Ser show highly favourable termini–termini interaction energy as compared to Pro. This may be due to the large size of Pro, which induces steric clashes hindering the residues to come close for interactions. This is also reflected in the positive osmotic coefficient of Pro indicating repulsive solute–solute interactions. Interestingly, the probability distributions of intermolecular energy (Fig. S10, ESI†) show that the termini–termini energy is favourable in the order Gly > Ala > Ser  $\approx$  Thr > Pro, which is also the increasing order of the size of the

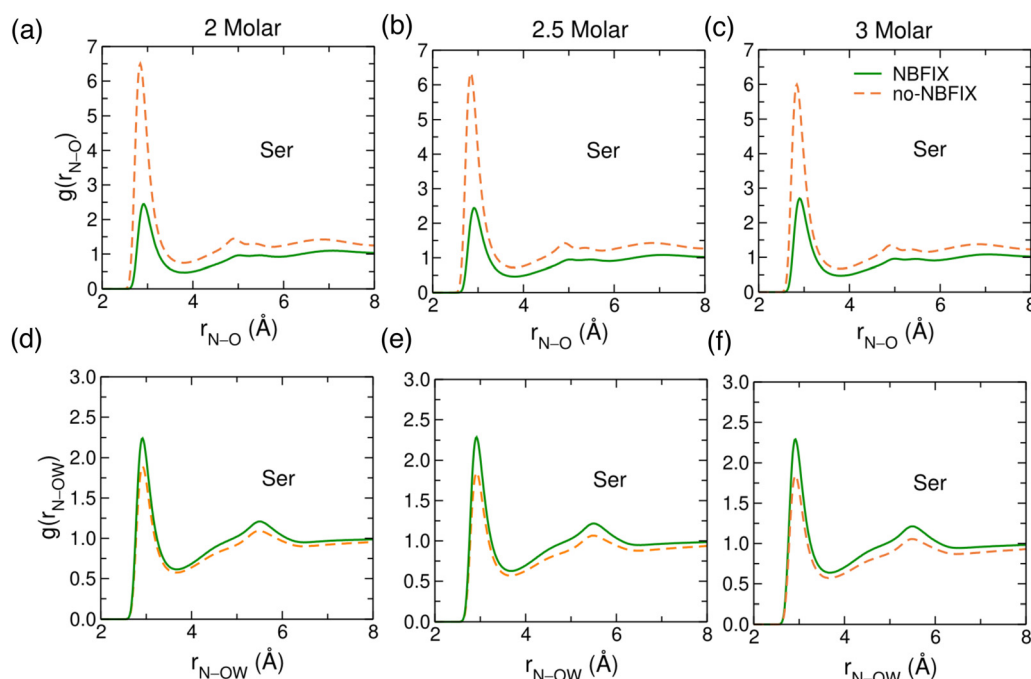


Fig. 6 Radial distribution functions of (a)–(c) solute–solute (amine nitrogen with carboxylate oxygen) intermolecular interactions and (d)–(f) solute–water (amine N with water oxygen) interactions for Ser solutions when original (no-NBFIX) and NBFIX parameterization are used. The three columns represent three concentrations, 2 M, 2.5 M and 3 M of the aqueous solutions.

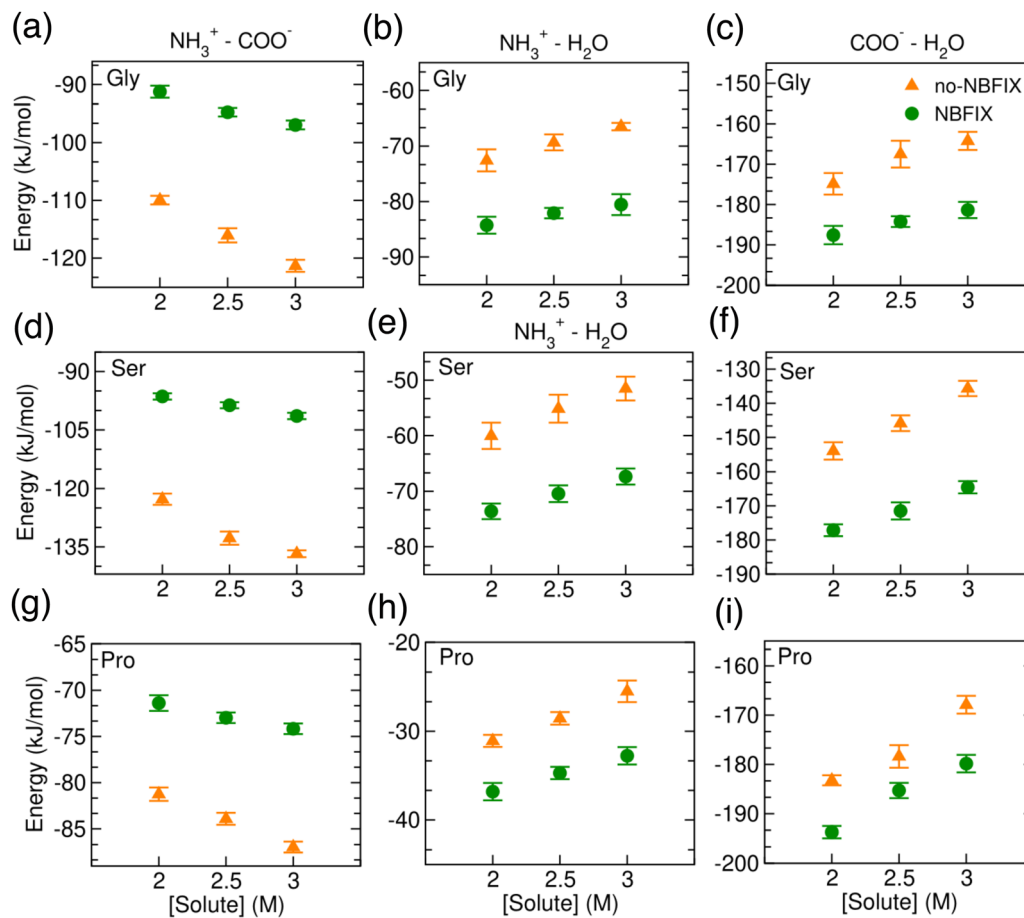


Fig. 7 Energy of interactions between (first column) the amine group and carboxylate group ( $\text{NH}_3^+ - \text{COO}^-$ ) of the solute zwitterionic pair, (second column) the amine group of the zwitterion and water ( $\text{NH}_3^+ - \text{H}_2\text{O}$ ), and (third column) the carboxyl group of the zwitterion and water ( $\text{COO}^- - \text{H}_2\text{O}$ ) of Gly (a)–(c), Ser (d)–(f) and Pro (g)–(i) aqueous solutions as a function of the solute concentration when original (no-NBFX) and NBFX parameterization are used.

amino acid residues. However, the water–termini interaction energy is equally favourable for Gly, Ala, and Pro except for the hydrophilic residues, Ser and Thr, which show a distribution over a wide range of energies. Applying NBFX increases the range of energies of the termini–water interactions for Ser and Thr. Therefore, this indicates that the NBFX reparameterization is successful in reducing the over-attractive solute–solute interactions at high concentrations.

Since the solute–solvent interactions become slightly more favourable on applying the NBFX reparameterization, it would be interesting to examine the effect on the tetrahedral hydrogen bonding network of water molecules, as quantified by the tetrahedral order  $q_{\text{tet}}$ , in these aqueous solutions. Fig. 8 shows that the normalized probability distribution of the tetrahedral order ( $P(q_{\text{tet}})$ ) of the pure water system is bimodal with a peak at  $q_{\text{tet}} = 0.5$  (less ordered water) and a shoulder at  $q_{\text{tet}} = 0.8$  (highly ordered water).<sup>62</sup> The  $P(q_{\text{tet}})$  of crowded amino acid solutions is very similar to that of pure water indicating that the ordering of water molecules is not affected by crowded solutes since the water molecules are able to form H-bonds with the polar atoms of the solutes. It can be seen that the distributions are very

similar when either the original force-field is applied or NBFX parameterization is applied (see Fig. S11, ESI†). This indicates that modifying the solute–solute interaction parameter does not affect the water–water interactions.

Packing the amino acids at high packing fractions is expected to affect the mobility of water. Fig. 9 shows the mean squared displacement (msd) of water as a function of time at 3 M concentration of amino acid solutions using NBFX parameterization. The increasing order of msd of water is Gly < Ala < Ser < Pro  $\approx$  Thr as shown in Fig. 9(a). Crowded environments are expected to induce anomalous diffusion of water.<sup>63,64</sup> The dependence of msd on time is given by  $\text{msd}(t) = Dt^{2/d_w}$ , where  $d_w$  is the anomalous diffusion exponent and  $D$  is the diffusion coefficient.<sup>63</sup> If  $d_w = 2$ , there is linear dependence of msd on time, which indicates normal diffusion of water. If  $d_w > 2$ , it indicates slowed-down, anomalous diffusion. The plots of  $\log(\text{msd}/t)$  vs.  $\log t$  can provide a better understanding of the anomalous diffusion as shown in Fig. 9(b). For  $t < 10$  ps, water shows anomalous diffusion in all the amino acid solutions at 3 M. Such an anomalous regime indicates caging of water molecules due to a large number of solute molecules

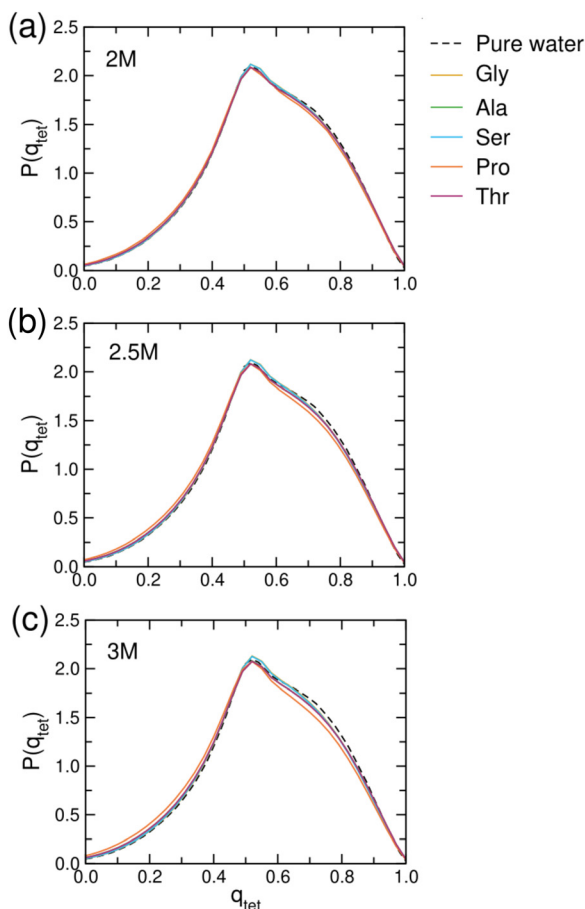


Fig. 8 Normalized probability distribution of the tetrahedral order parameter of water ( $P(q_{\text{tet}})$ ) in aqueous solutions of amino acids with NBFIX reparameterization. The dotted lines represent the  $P(q_{\text{tet}})$  of the pure bulk water.

acting like obstacles at high packing fractions. At larger time scales,  $t > 10$  ps, the diffusion becomes normal (see Fig. S12, ESI† for details). **With increasing size of the amino acid, the anomalous diffusion regime increases in the order: Gly < Ala < Ser < Pro ≈ Thr.** For a given amino acid solution, the cross-over time from anomalous to normal diffusion increases as the crowding in the solution increases. The anomalous diffusion regimes are also observed at 2 M and 2.5 M for all the amino acids (see Fig. S13, ESI†) and the msd decreases with increasing concentration for a given amino acid. This is also reflected in the concentration-dependence of the diffusion coefficient of water,  $D_w$ , as shown in Fig. 9(c) where  $D_w$  decreases with increasing crowding and decreases at a given concentration with the increasing size of the amino acid. On comparing no-NBFIX and NBFIX parameterization, it is found that the differences are not significant in the msd of water in all the amino acid solutions, except for Ser and Thr, where the difference in the cross-over time is observed at 3 M (see Fig. S13, ESI†). With no-NBFIX, the cross-over from anomalous to normal diffusion occurs at a longer time as compared to when NBFIX is used. This may be due to a higher number of water–water hydrogen bonds when no-NBFIX is used due to which the decay in the

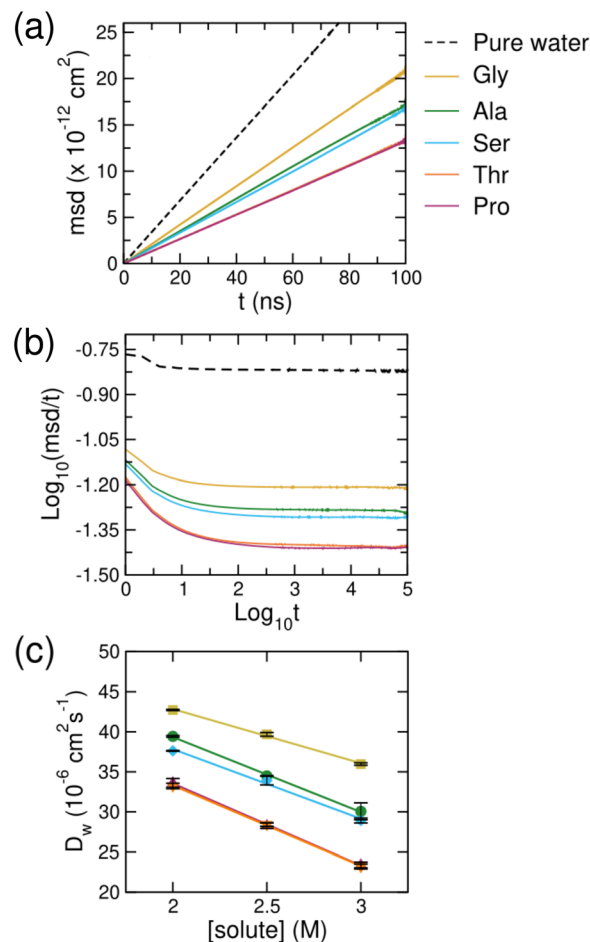


Fig. 9 (a and b) Mean squared displacement (msd) of water molecules as a function of time in amino acid solutions at 3 M. (c) Diffusion coefficient of water as a function of amino acid concentration using NBFIX reparameterization.

$\log(\text{msd}/t)$  is slower. The tendency of water to form hydrogen bonds with the amino acids gets increased when NBFIX is applied, as reflected in the favourable solute–water interaction energy (Fig. 8). As a result, the number of water–water hydrogen bonds gets reduced due to which the cross-over to normal diffusion is faster in the case of NBFIX.

## Discussion and conclusions

We investigate the performance of modified AMBER ff99sb-ILDN with a TIP3P water model at high concentrations of Ala, Gly, Ser, Thr and Pro amino acids corresponding to the crowded packing fractions. We follow the reparameterization strategy proposed by Yoo *et al.*, which involves incrementing the Lennard-Jones  $\sigma$  parameter (by  $\Delta\sigma$ ) for the intermolecular interaction of amine N and carboxyl O of the zwitterionic termini of amino acids.<sup>40</sup> **Miller *et al.* have assessed the performance of this reparameterization for reproducing experimental osmotic coefficients of amino acids at 2 M concentration.**<sup>41</sup> **However, such a reparameterization strategy has not been assessed for computing the bulk solution**

properties of amino acid solutions at higher packing fractions or crowded concentrations, i.e. above 2 M. We perform a systematic quantitative investigation of bulk thermodynamic and dynamic properties of these aqueous solutions at crowded concentrations. Our results show that the modified force-field (NBFIX) parameterization improves the agreement of simulated osmotic coefficients with the experimental values at the crowded concentrations. Additionally, we test the performance of other values of  $\Delta\sigma$  and find that using a lower value of  $\Delta\sigma$  at the highest concentration studied, reduces the overall deviation of simulated values of  $\phi$  with the experimental values for all the amino acids. The modified force-field also performs well in reproducing the concentration-dependence of the viscosity of solutions. **The modified force-field performs reasonably well in reducing the deviation from experiments of the concentration-dependence of the self-diffusivity coefficients of amino acids. However, both the original and modified parameterization reproduce the experimental densities of crowded aqueous solutions equally well.**

We further examine the hydration structure, energetics and dynamics in these crowded solutions. We find that the solute–solute interactions become less favourable as reflected in the radial distribution functions and the energy of interaction. Correspondingly, the termini–water interaction energy becomes more favourable indicating reduction in the aggregation of these solutes at such high concentrations. **NBFIX parameterization is also found to improve the convergence of the tails of rdfs for solute–solute and solute–water interactions, implying the decrease in the finite-size effects.** However, reparameterizing the termini–termini interactions does not significantly affect the structuring of water in the first hydration shell, as quantified by the tetrahedral order. **Interestingly, water shows anomalous diffusion at small time scales in these crowded solutions before switching to normal diffusion at larger time scales, as was also observed in protein-crowded solutions in a previous study.<sup>64</sup> Our results show that the diffusion coefficient of water decreases with increasing amino acid concentration and amongst the different aqueous solutions, it is primarily affected by the various sizes of the amino acids, consistent with the trends observed for self-diffusivity of amino acids.** Moreover, the anomalous diffusion of water is found to be sensitive to the type of amino acid solution being studied. **This implies that the dynamics of water in crowded solutions would be sensitive to the sequence of the proteins. This is expected to have significant consequences on the structure and dynamics of biomolecules and needs to be explored further.**

Miller *et al.* have shown the improvement in performance of other force-fields such as CHARMM36,<sup>23</sup> GROMOS54a7,<sup>24,65</sup> and OPLS-AA,<sup>66</sup> after applying the NBFIX parameterisation to amino acid solutions at 2 M concentration.<sup>41</sup> We, therefore, expected that our results for NBFIX reparameterization at high solute concentrations would perform in a similar way with the other force-fields as well; however, a systematic comparison constitutes the scope of a future study. Due to the lack of available data for osmotic pressure at high concentrations,

we considered only five amino acids in this work. The results of this study should be applicable to all the amino acids except for the charged ones like Glu, His, and Lys where it has been shown that the original force-field is capable of producing  $\phi$  values close to experiments.<sup>41</sup> However, it would be interesting to explore the performance of such reparameterization for these solutions at different pH values where the counterion (salt)–water interactions would play a crucial role.<sup>47,67</sup> The transferrability of NBFIX parameters to other kinds of molecules with different chemical nature or larger sizes such as oligopeptides would require additional examination and validation.<sup>42</sup> We tested the application of NBFIX to AMBER ff99sb-ILDN to match the experimental osmotic coefficients for diglycine peptide (see Fig. S14, ESI†) at 2 M, 2.5 M and 3 M crowded concentrations. The results show that there are large differences between the experimental and the simulated  $\phi$  values using the original AMBER ff99sb-ILDN force-field. Even after using modified partial charges of the dipeptides<sup>47,68</sup> and testing the NBFIX reparameterization with a higher value of  $\Delta\sigma = 0.16$  Å, the simulated  $\phi$  values did not match the experimental values at and above 2 M. It would be, therefore, interesting to assess the modification in solute termini–water LJ interactions as suggested by Nerenberg *et al.*<sup>35</sup> in addition to the NBFIX parameterization to obtain accurate osmotic coefficients at high concentrations of dipeptides. Other rescaling techniques such as rescaling the electrostatic partial charges on solute atoms, which have been adopted in the literature for other small solutes, could be another plausible route to improve the thermodynamic properties.<sup>48,69</sup> It would also be interesting to employ machine learning algorithms to improve the non-bonded interaction parameters for a given type of oligopeptide (or a small protein), which would help in eliminating the artificial aggregation in such solutions. However, for small molecules such as the small-sized alcohols, we expect that the NBFIX parameterization would work at crowded concentrations, as shown by Miller *et al.* for low concentrations of aliphatic alcohols.<sup>41</sup> Our results, therefore, have implications for modeling other molecular crowded systems such as of ethylene glycol and triethylene glycol. The anomalous dynamics of water in such crowded solutions, as highlighted by our findings, implies that accounting for crowding in examining biomolecular processes is of utmost importance. It would be interesting to explore the effect of differences in the properties of water models on reproducing the thermodynamic properties and the anomalous behaviour of water in crowded solutions. For instance, we find that using the TIP4P/2005 water model with NBFIX reparameterized AMBER ff99sb-ILDN produces an osmotic coefficient of 3 M Gly solution to be 1.36 ( $\pm 0.016$ ) in comparison to the original force-field producing the value of 1.15 ( $\pm 0.020$ ), with the experimental value being 0.88.<sup>51</sup> These obtained values are different from those obtained with the TIP3P water model (see Fig. 2). This implies that a systematic investigation is needed before applying the NBFIX strategy when using different water force fields. Additionally, the improvement in the convergence of the tails observed for solute–solute radial distribution functions on application of NBFIX circumvents the

issue of finite-size effects to some extent and would be expected to improve the convergence of Kirkwood–Buff (KB) integrals.<sup>61,70</sup> Given the current research going on in developing KB force-fields for proteins,<sup>71</sup> it makes it an interesting direction to explore for obtaining accurate thermodynamic properties of crowded aqueous solutions.

## Author contributions

S. P. and R. S. contributed to the investigation, data curation, and formal analysis and contributed equally to the work. S. K. R. contributed to the methodology, funding acquisition, project administration, supervision, and writing – review and editing. D. N. contributed to the conceptualization, methodology, funding acquisition, project administration, supervision, and writing – original draft.

## Conflicts of interest

The authors declare no competing conflicts of interest.

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